

Algebra 2: Unit 6 Day 1 — ANSWER KEY

Part 1: Basic Isolation

1. $\sqrt[3]{x} = 5 \rightarrow x = 5^3$

Answer: 125

2. $2\sqrt[3]{x-5} = -4 \rightarrow \sqrt[3]{x-5} = -2$
 $x - 5 = -8$

Answer: -3

3. $\sqrt[3]{x} + 8 = 10 \rightarrow \sqrt[3]{x} = 2$

Answer: 8

4. $\sqrt[3]{x+1} = 4 \rightarrow x+1 = 64$

Answer: 63

5. $\frac{\sqrt[3]{x}}{2} = 3 \rightarrow \sqrt[3]{x} = 6$

Answer: 216

6. $\sqrt[3]{x} - 7 = -10 \rightarrow \sqrt[3]{x} = -3$

Answer: -27

Part 2: Rational Exponents

1. $x^{1/3} = 4 \rightarrow x = 4^3$

Answer: 64

2. $x^{2/3} = 25 \rightarrow x = (25)^{3/2} \rightarrow (\sqrt{25})^3$

Answer: 125

Part 3: One-Step Operations

1. $\sqrt[3]{x+7} = 3 \rightarrow x+7 = 27$

Answer: 20

2. $\sqrt[3]{2x} = 4 \rightarrow 2x = 64$

Answer: 32

3. $(x+1)^{2/3} = 4 \rightarrow x+1 = (4)^{3/2} = 8$

Answer: 7

4. $(x-4)^{2/3} = 16 \rightarrow x-4 = (16)^{3/2} = 64$

Answer: 68

You Try!

1. $\sqrt[3]{x} = -2 \rightarrow x = (-2)^3$

Answer: -8

2. $x^{1/3} + 5 = 8 \rightarrow x^{1/3} = 3 \rightarrow x = 3^3$

Answer: 27

3. $3\sqrt[3]{x} = 15 \rightarrow \sqrt[3]{x} = 5$

Answer: 125

4. $x^{2/3} = 49 \rightarrow x = (49)^{3/2} = 7^3$

Answer: 343

5. $\sqrt[3]{x-10} = -3 \rightarrow x-10 = -27$

Answer: -17

6. $(x+5)^{2/3} = 1 \rightarrow x+5 = 1^{3/2} = 1$

Answer: -4 (also -6)

7. $\sqrt[3]{x+2} = 5 \rightarrow x+2 = 125$

Answer: 123

8. $2\sqrt[3]{x} = 12 \rightarrow \sqrt[3]{x} = 6$

Answer: 216

9. $(x-2)^{1/3} = -4 \rightarrow x-2 = -64$

Answer: -62

10. $\sqrt[3]{3x} = 3 \rightarrow 3x = 27$

Answer: 9